

## Day 11: Inference for MLR

MLR model for `bikes` data

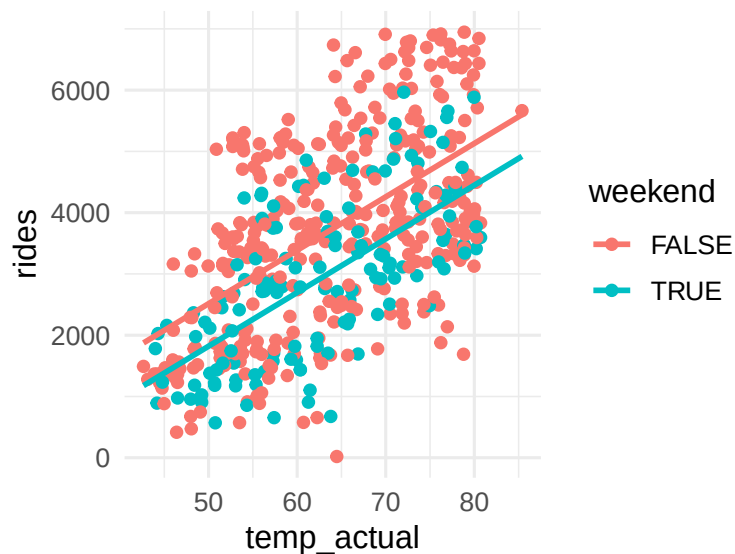
We're back to the separate-lines model from last class:

| term                    | estimate  | std. error | t     | p-value |
|-------------------------|-----------|------------|-------|---------|
| Intercept               | -1843.867 | 415.700    | -4.44 | <0.001  |
| temp_actual             | 87.173    | 6.439      | 13.54 | <0.001  |
| weekendTRUE             | -714.938  | 795.698    | -0.90 | 0.369   |
| temp_actual:weekendTRUE | 0.411     | 12.539     | 0.03  | 0.974   |

### 💡 Warm up

Write out the theoretical and fitted regression model for `rides ~ temp_actual * weekend`.

Do we need a different-slopes model?



Look at the `temp_actual : weekendTRUE` row in the table above – its estimate is tiny and its p-value is large, suggesting the slopes might not really differ.

## Testing a single coefficient

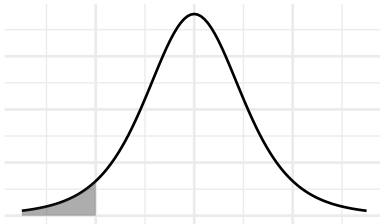
**Hypotheses:**  $H_0 : \beta_j = \#$  vs.  $H_a : \beta_j \neq, <, \text{ or } > \#$

**Test statistic:**  $t = \frac{\hat{\beta}_j - \#}{SE(\hat{\beta}_j)}$

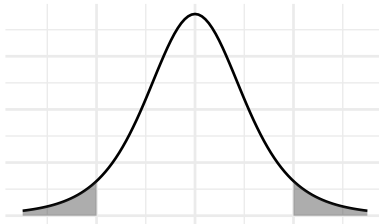
**Reference distribution:**  $t$  distribution with  $df = n - (k + 1)$ , where  $k$  = number of predictors in the model

**p-value:** area in the tail(s) specified by  $H_a$

$H_a : \beta_j < 0$



$H_a : \beta_j \neq 0$



$H_a : \beta_j > 0$



**CI for a single coefficient:**  $\hat{\beta}_j \pm t_{n-(k+1)}^* \cdot SE(\hat{\beta}_j)$

## Do we need weekend at all?

Careful – with the interaction term still in the model, the weekendTRUE row alone doesn't answer “do we need weekend at all,” because weekend also appears in the interaction term.

| term                    | estimate  | std. error | t     | p-value |
|-------------------------|-----------|------------|-------|---------|
| Intercept               | -1843.867 | 415.700    | -4.44 | <0.001  |
| temp_actual             | 87.173    | 6.439      | 13.54 | <0.001  |
| weekendTRUE             | -714.938  | 795.698    | -0.90 | 0.369   |
| temp_actual:weekendTRUE | 0.411     | 12.539     | 0.03  | 0.974   |

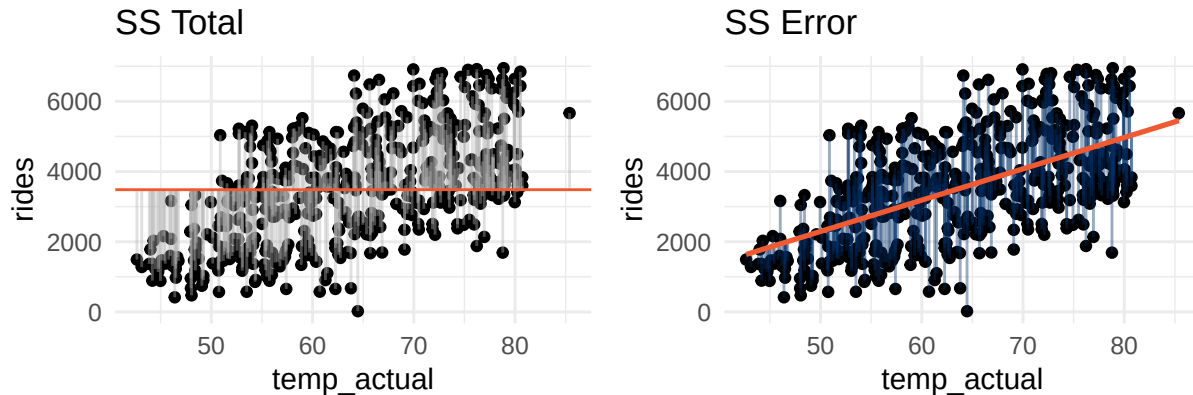
### 🔥 Caution!

A single coefficient's p-value only tests that \_\_\_ term, holding all others in the model. To ask whether weekend matters *at all*, we need to test  $\beta_2 = \beta_3 = 0$  jointly – more on this below.

Once we drop the (non-significant) interaction, the simpler parallel-lines model shows weekend is clearly still useful:

| term        | estimate  | std. error | t     | p-value |
|-------------|-----------|------------|-------|---------|
| Intercept   | -1850.776 | 357.953    | -5.17 | <0.001  |
| temp_actual | 87.281    | 5.520      | 15.81 | <0.001  |
| weekendTRUE | -689.179  | 124.639    | -5.53 | <0.001  |

Explained variability



**Idea:** does a larger model explain a *significantly* bigger share of the total variability than a smaller model?

Can we just compare  $R^2$  values?

| Model                      | $R^2$  | Adjusted $R^2$ |
|----------------------------|--------|----------------|
| temp only                  | 0.3322 | 0.3308         |
| temp + weekend             | 0.3709 | 0.3683         |
| temp + windspeed + weekend | 0.3898 | 0.3861         |
| temp + windspeed*weekend   | 0.3901 | 0.3852         |

$$R^2 = 1 - \frac{SSE}{SST} \quad R^2_{adj} = 1 - \frac{SSE/(n-k)}{SST/(n-1)} = 1 - \frac{MSE}{MST}$$

- $R^2$  **always** increases when predictors are added, even useless ones
- $R^2_{adj}$  **could** \_\_\_\_\_ if unnecessary predictors are added
- Better to use  $R^2_{adj}$  for comparing multiple regression models

Formal test for comparing models

|         | Model                                                            |
|---------|------------------------------------------------------------------|
| Full    | $\mu\{Y X\} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1x_2$ |
| Reduced | $\mu\{Y X\} = \beta_0 + \beta_1x_1$                              |

**Hypotheses:**  $H_0 : \beta_2 = \beta_3 = 0$  vs.  $H_a : \text{at least one } \beta_j \neq 0, j = 2, 3$

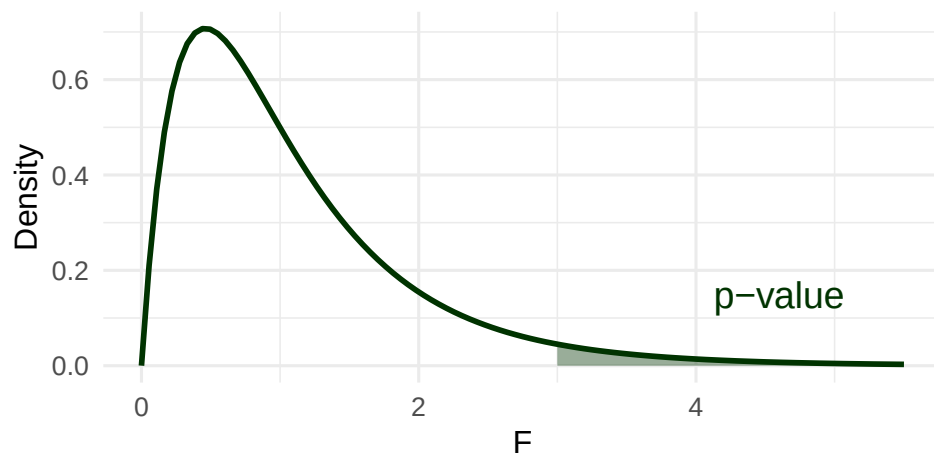
Comparing models with an F-test

$$F = \frac{(R_{\text{full}}^2 - R_{\text{reduced}}^2)/d}{(1 - R_{\text{full}}^2)/df_{\text{full}}} = \frac{(SSR_{\text{full}} - SSR_{\text{reduced}})/d}{MSE_{\text{full}}} = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/d}{MSE_{\text{full}}}$$

where

- $d = df_{\text{full}} - df_{\text{reduced}} = \# \text{ of } \beta\text{'s being tested}$
- $df_i = n - (p + 1) = \text{error d.f. for model } i$
- $MSE_{\text{full}} = \frac{SSE_{\text{full}}}{df_{\text{full}}}$

The  $F$  statistic follows an  $F$  distribution with  $d$  and  $n - p - 1$  degrees of freedom. Like the  $\chi^2$ , it's right-skewed and only takes positive values, so we only ever look at the **upper tail** for a p-value:



Upper-tail area in R: `1 - pf(stat, df1, df2)`

Putting it all together

| model              | r.squared | df | df.residual | nobs |
|--------------------|-----------|----|-------------|------|
| Full: temp*weekend | 0.371     | 3  | 496         | 500  |
| Reduced: temp only | 0.332     | 1  | 498         | 500  |

$$F = \frac{(R_{\text{full}}^2 - R_{\text{reduced}}^2)/d}{(1 - R_{\text{full}}^2)/df_{\text{full}}} = \frac{(0.371 - 0.332)/2}{(1 - 0.371)/496} \approx 15.3$$

p-value = 3.71e-07 – extremely small, so we \_\_\_\_\_  $H_0$ : adding weekend (and its interaction with temp\_actual) meaningfully improves the model.

Doing it directly in R with `anova()`

`anova(reduced, full)` takes the **reduced** model first, then the **full** model, and runs exactly the F-test above – no need to compute  $R^2$ 's by hand.

Analysis of Variance Table

Model 1: rides ~ temp\_actual

Model 2: rides ~ temp\_actual \* weekend

|   | Res.Df | RSS       | Df | Sum of Sq | F      | Pr(>F)        |
|---|--------|-----------|----|-----------|--------|---------------|
| 1 | 498    | 826326178 |    |           |        |               |
| 2 | 496    | 778436469 | 2  | 47889709  | 15.257 | 3.713e-07 *** |

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